PULLBACK ON THE SECOND COHOMOLOGY WITH ROOTS OF UNITY IS MULTIPLICATION BY THE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the closing 'Some details omitted' refers to the verification of naturality of the Kummer coboundary, a routine compatibility, and it is preserved verbatim in the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The second cohomology of a smooth projective curve with roots of unity. Same chapter, Lemma lemma-cohomology-smooth-projective-curve, which identifies both groups with Z/nZ. Gives the identification with which the statement makes sense.
- The Kummer sequence and its coboundary. Same chapter, the long exact cohomology sequence of the Kummer sequence. Produces the generator of the second cohomology group as the coboundary of the class of a line bundle.

Source reference. Original lemma 03RQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11135–11135. Navigation pointer invoked at proof line 11234; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\pi : X \to Y$ be a nonconstant morphism of smooth projective curves over an algebraically closed field k and let $n \geq 1$ be invertible in k. The map*

$$\pi^* : H^2_{\acute{e}tale}(Y, \mu_n) \longrightarrow H^2_{\acute{e}tale}(X, \mu_n)$$

is given by multiplication by the degree of π.

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Proof. Observe that the statement makes sense as we have identified both cohomology groups $H^2_{\acute{e}tale}(Y, \mu_n)$ and $H^2_{\acute{e}tale}(X, \mu_n)$ with $\mathbf{Z}/n\mathbf{Z}$ in Lemma 03RQ. In fact, if $\mathcal{L}$ is a line bundle of degree 1 on Y with class $[\mathcal{L}] \in H^1_{\acute{e}tale}(Y, \mathbf{G}_m)$, then the coboundary of $[\mathcal{L}]$ is the generator of $H^2_{\acute{e}tale}(Y, \mu_n)$. Here the coboundary is the coboundary of the long exact sequence of cohomology associated to the Kummer sequence. Thus the result of the lemma follows from the fact that the degree of the line bundle $\pi^*\mathcal{L}$ on X is $\deg(\pi)$. Some details omitted. $\square$